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MIL-M-28787/346
26 October 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

INTERRUPT CONTROL UNIT MODULE, KEY CODE SQY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interrupt control unit module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 1.2882 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be SQY.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 130 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 + 5°C 4/		0 + 5°C 4/		60 + 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation,	II								
	initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
A2-A54	Isolation,	II								
	initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
B1-B4	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B5-B8	Continuity	III	Q	0.0	0.1	0 0	0.1	0.0	0.1	Ohm
C	Vcc stress	IV	Q	----------Module shall pass all remaining tests----------						
D1	Ibd	V	S	0.0	1.1	0 0	1.1	0.0	1.1	mA
D2-D11	Ibd	V	S	0.0	1.0	0 0	1.0	0.0	1.0	mA
D12-D27	Ibd	V	S	0.0	0.13	0.0	0.13	0.0	0.13	mA
D28-D37	Ibd	V	S	0.0	0.1	0 0	0.1	0.0	0.1	mA
E1-E37	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
F1	Ios	VII	S	-130	-30	-130	-30	-130	-30	mA
F2-F11	Ios	VII	S	-225	-40	-225	-40	-225	-40	mA
G1	Iih	VIII	X	0	45	0	45	0	45	uA
G2-G11	Iih	VIII	X	0	25	0	25	0	25	uA
G12-G27	Iih	VIII	X	-887	-783	-887	-783	-887	-783	uA
G28,G29	Iih	VIII	X	0	20	0	20	0	20	uA
G30-G37	Iih	VIII	X	-20	20	-20	20	-20	20	uA
H1	Iil	IX	X	-450	-5	-450	-5	-450	-5	uA
H2-H11	Iil	IX	X	-250	0	-250	0	-250	0	uA
H12-H27	Iil	IX	X	-1817	-1468	-1817	-1468	-1817	-1468	uA
H28,H29	Iil	IX	X	-400	0	-400	0	-400	0	uA
H30-H37	Iil	IX	X	-200	-5	-200	-5	-200	-5	uA
J1	Voh	X	X	2.4	2 6	2 4	2.6	2.4	2.6	V
J2-J12	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
K1	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K2-K11	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
L1-L8	Iozh	XII	X	-20	20	-20	20	-20	20	uA
M1-M8	Iozl	XIII	X	-200	20	-200	20	-200	20	uA
N	Icc	XIV	X	70	754	70	754	70	754	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R	PSFC	XVIII	S	3.7	6.9	3.7	6.9	3.7	6.9	uF
S	PSTA	XIX	Q	0	400	0	400	0	400	mV
T1-T10	Output crosstalk (high level)	XX	Q	0	400	0	400	0	400	mV
T11	Output crosstalk (high level)	XX	Q	0	500	0	500	0	500	mV
U1-U10	Output crosstalk (low level)	XXI	Q	0	400	0	400	0	400	mV
U11	Output crosstalk (low level)	XXI	Q	0	500	0	500	0	500	mV

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
V1	Cin	XXII	Q	4	90	4	90	4	90	pF
V2, V3	Cin	XXII	Q	2	15	2	15	2	15	pF
V4	Cin	XXII	Q	2	20	2	20	2	20	pF
V5	Cin	XXII	Q	2	30	2	30	2	30	pF
V6, V7	Cin	XXII	Q	2	25	2	25	2	25	pF
V8	Cin	XXII	Q	2	20	2	20	2	20	pF
V9-V11	Cin	XXII	Q	2	15	2	15	2	15	pF
V12-V27	Cin	XXII	Q	2	40	2	40	2	40	pF
V28, V29	Cin	XXII	Q	2	15	2	15	2	15	pF
W1, W2	Cout	XXIII	Q	2	20	2	20	2	20	pF
W3	Cout	XXIII	Q	2	25	2	25	2	25	pF
W4-W11	Cout	XXIII	Q	2	55	2	55	2	55	pF
X	Threshold	XXIV	X	----------		----------	See table XXIV	----------		
Y1-Y16	Disable time	XXV	Q	2	84	2	84	2	84	nS

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/346-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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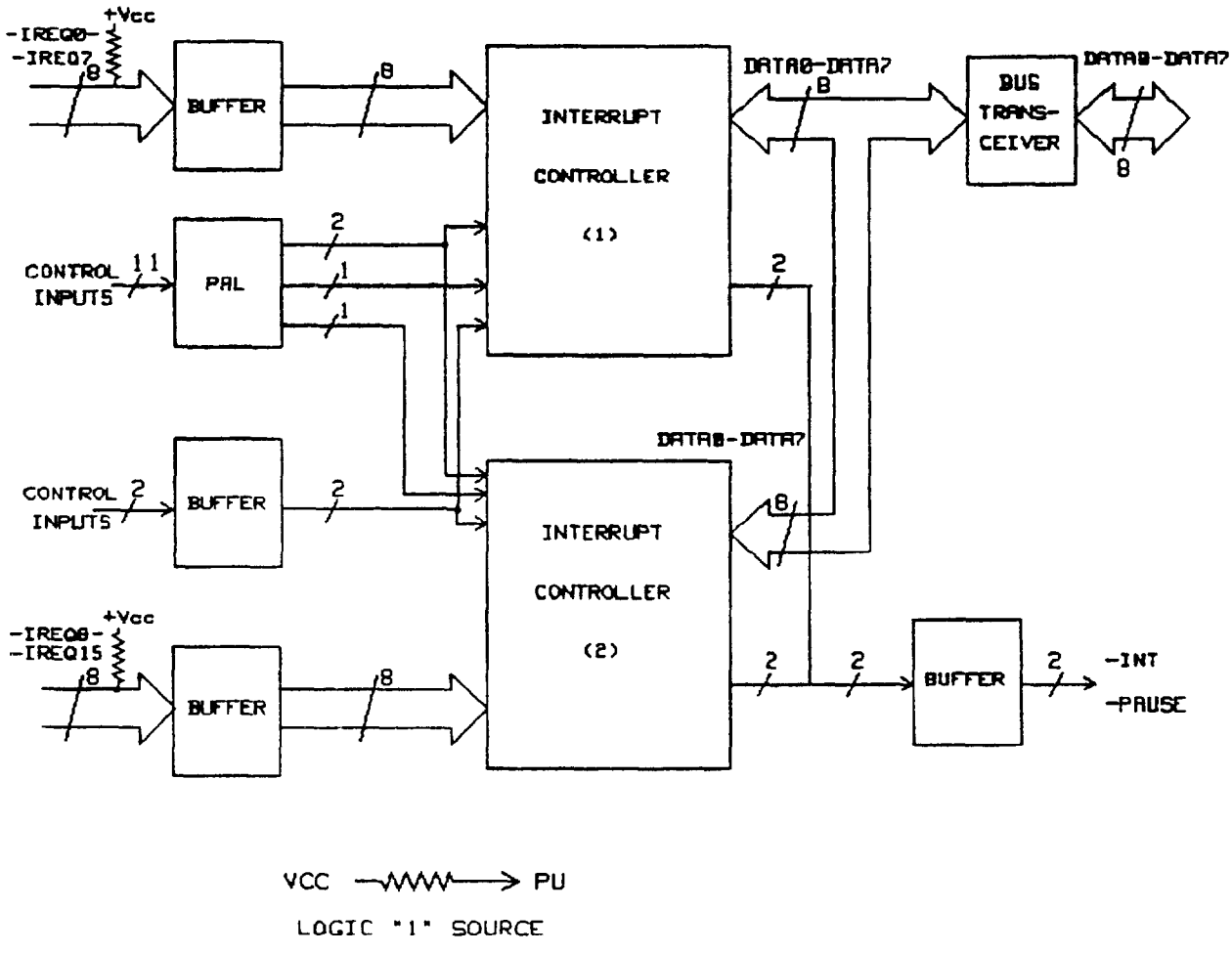


FIGURE 1. SQY functional block diagram.

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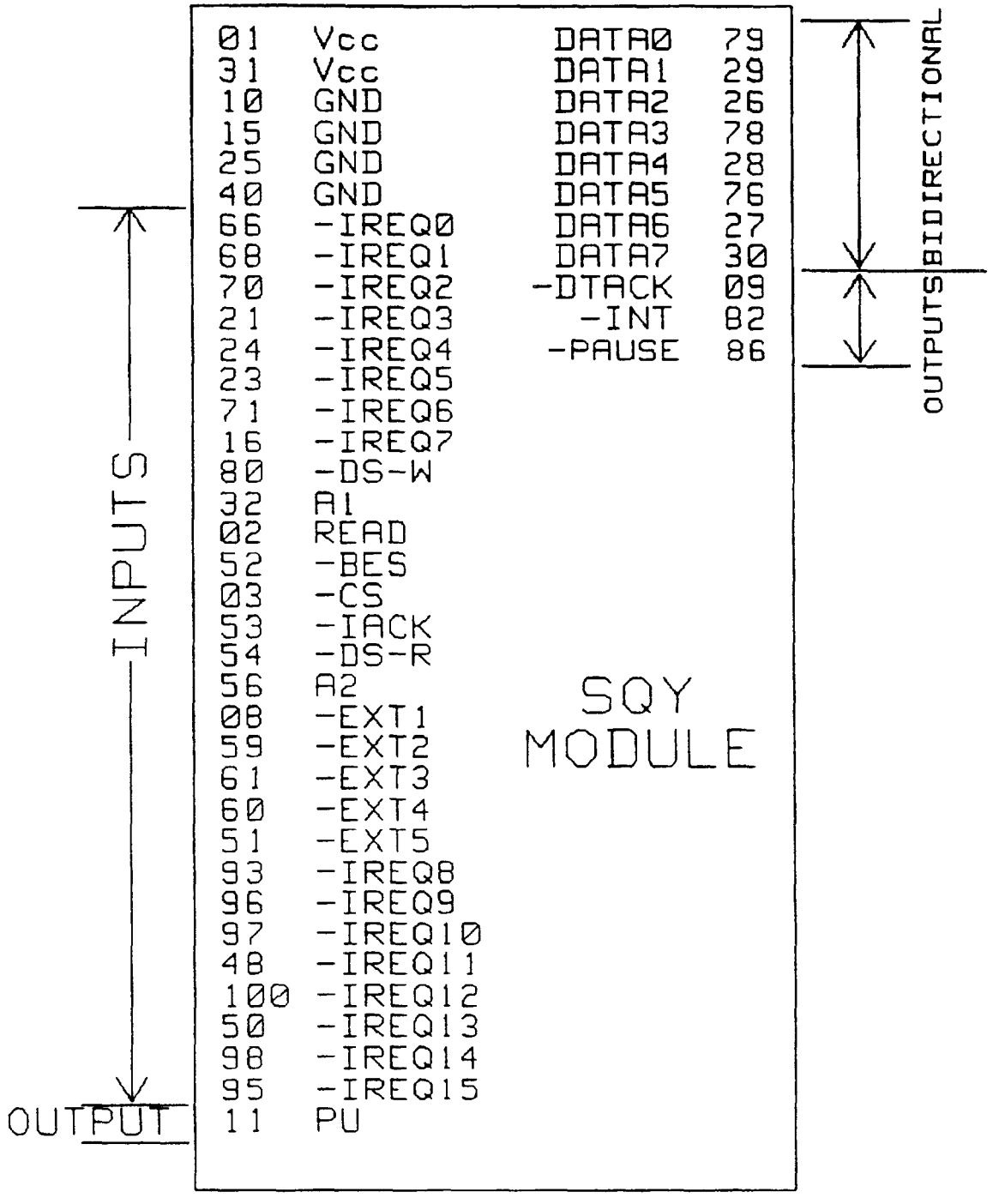


FIGURE 2. SQY pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

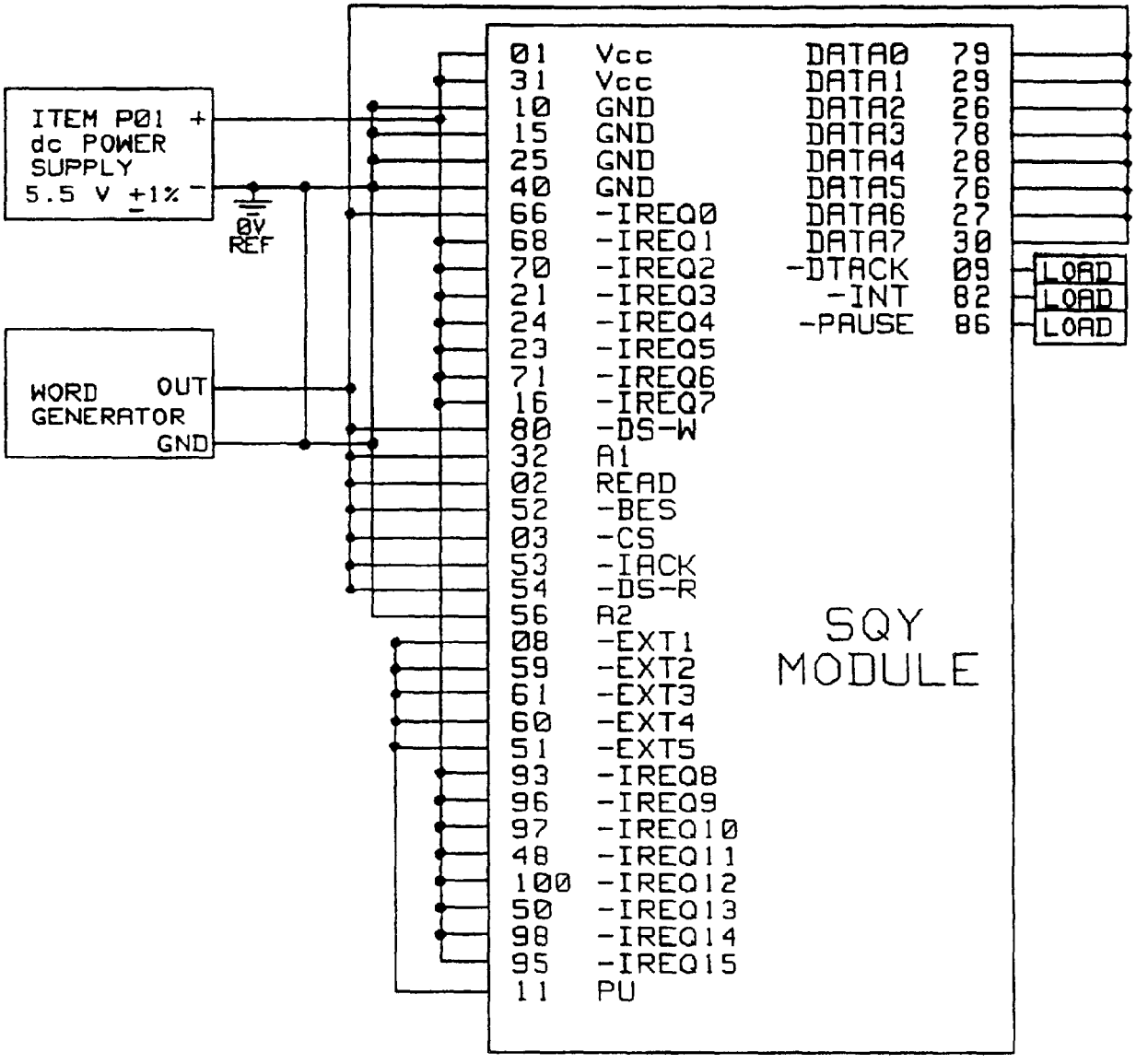
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through U, X and Y shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 4 and 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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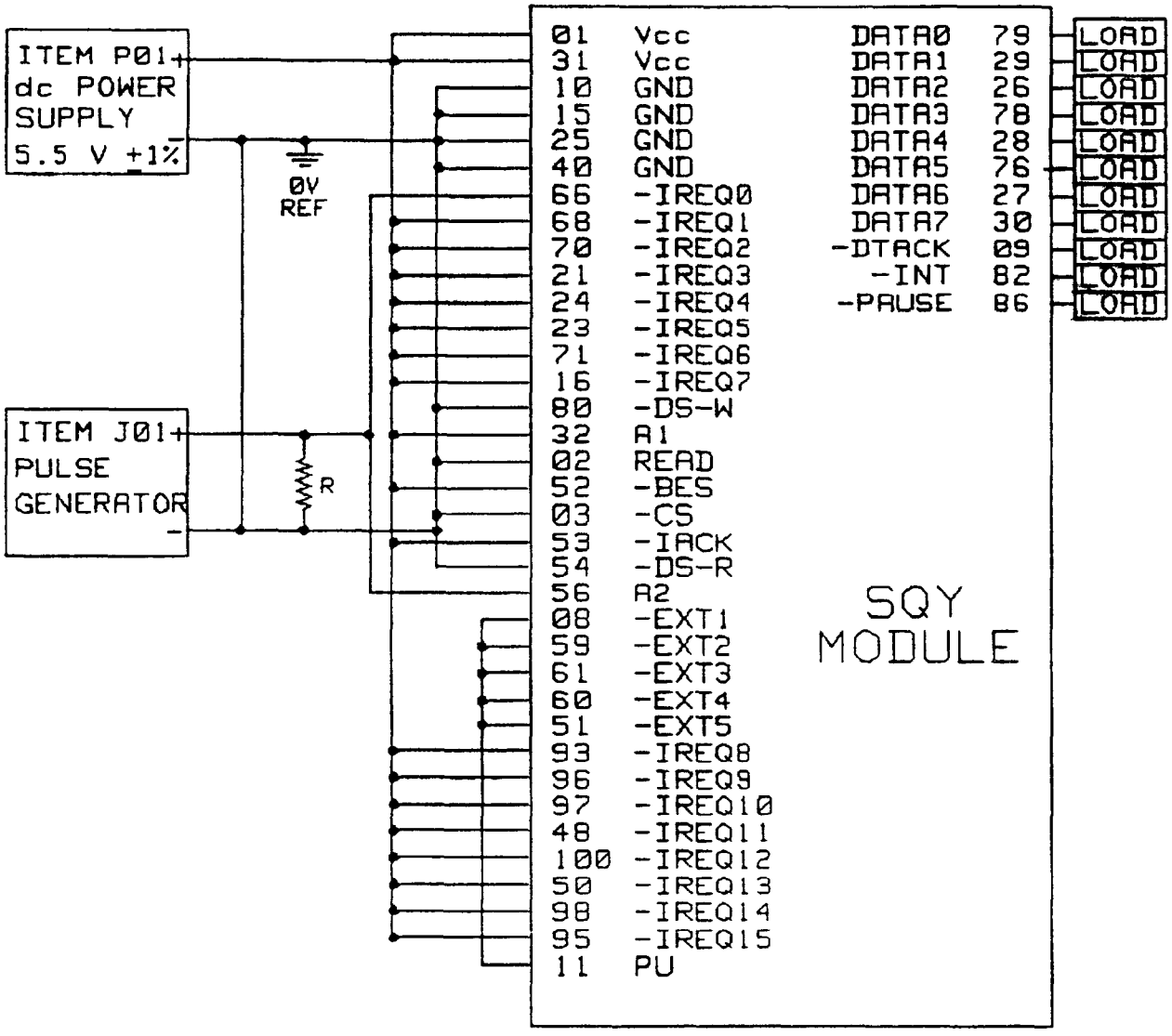
NOTES:

1. Load outputs as specified in the output voltage test.

2. Pattern voltages are as follows: High level (Vih) = 3.0 V ± 0.1 V; low level (Vil) = 0.2 V ± 0.1 V.

FIGURE 3. Operating temperature and life test circuit.

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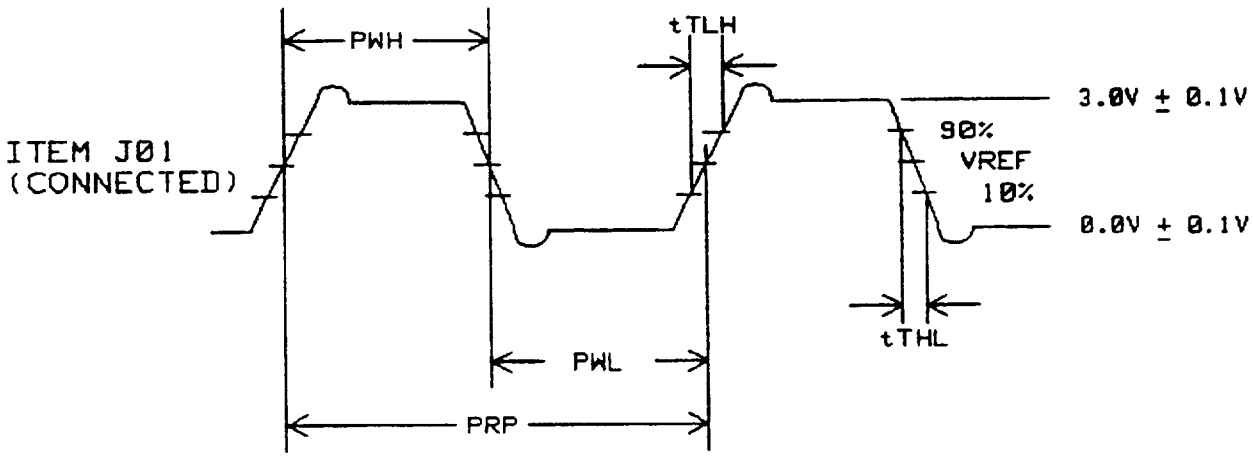


NOTES:

1. R = 51 Ohms ± 5 percent, 1/2 W.
2. Load outputs as specified in the output voltage test.

FIGURE 4. Component temperature test circuit.

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NOTES:

1. PWH = 300 ± 20 nS.
2. PWL = 300 ± 20 nS.
3. PRP = 600 ± 40 nS.
4. tTLH = tTHL = 50 ± 10 nS.
5. VREF = 1.3 ± 0.1 V.

FIGURE 5. Component temperature test waveforms.

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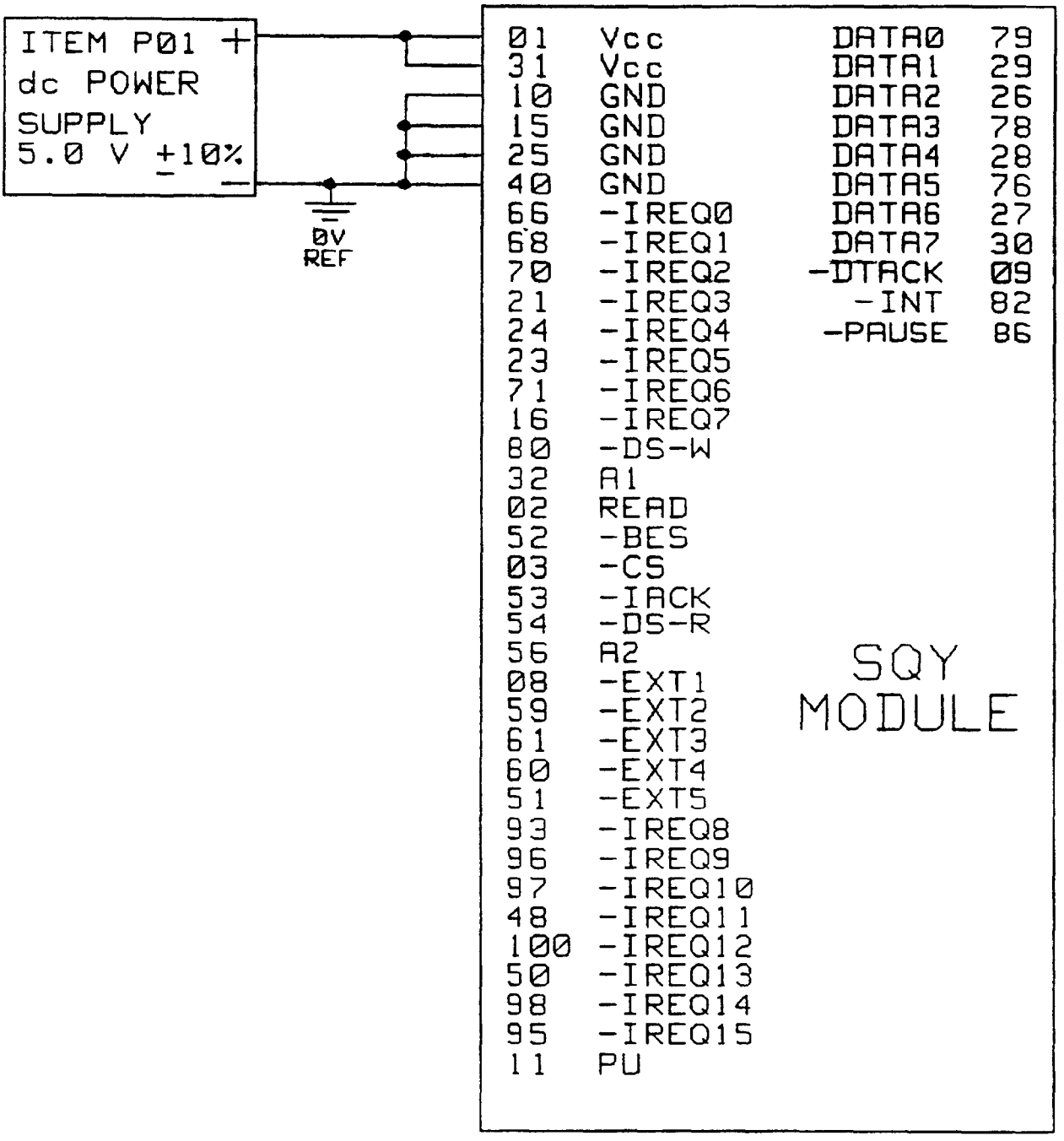


FIGURE 6. Vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests B, G through U, X and Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through U, X and Y shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 4.145 W (max); 2.23 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
-IREQ0	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ1	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ2	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ3	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ4	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ5	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ6	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ7	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ8	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ9	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ10	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ11	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ12	-1.5	0.7	2.0	-1817	-783	110	40

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
-IREQ13	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ14	-1.5	0.7	2.0	-1817	-783	110	40
-IREQ15	-1.5	0.7	2.0	-1817	-783	110	40
READ	-1.5	0.8	2.0	-450	45	1100	75
-BES	-1.5	0.8	2.0	-250	25	1000	20
-CS	-1.5	0.8	2.0	-250	25	1000	15
-IACK	-1.5	0.8	2.0	-250	25	1000	15
-DS-R	-1.5	0.8	2.0	-250	25	1000	15
A2	-1.5	0.8	2.0	-250	25	1000	15
-EXT1	-1.5	0.8	2.0	-250	25	1000	15
-EXT2	-1.5	0.8	2.0	-250	25	1000	20
-EXT3	-1.5	0.8	2.0	-250	25	1000	30
-EXT4	-1.5	0.8	2.0	-250	25	1000	25

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
-EXT5	-1.5	0.8	2.0	-250	25	1000	25
DATA0	-1.5	0.7	2.0	-200	20	100	55
DATA1	-1.5	0.7	2.0	-200	20	100	55
DATA2	-1.5	0.7	2.0	-200	20	100	55
DATA3	-1.5	0.7	2.0	-200	20	100	55
DATA4	-1.5	0.7	2.0	-200	20	100	55
DATA5	-1.5	0.7	2.0	-200	20	100	55
DATA6	-1.5	0.7	2.0	-200	20	100	55
DATA7	-1.5	0.7	2.0	-200	20	100	55
-DS-W	-1.5	0.7	2.0	-400	20	100	15
A1	-1.5	0.7	2.0	-400	20	100	15

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
DATA0	2.4	0.4	-3000	12.0	20.0	-200	-40.0	55
DATA1	2.4	0.4	-3000	12.0	20.0	-200	-40.0	55
DATA2	2.4	0.4	-3000	12.0	20.0	-200	-40.0	55
DATA3	2.4	0.4	-3000	12.0	20.0	-200	-40.0	55
DATA4	2.4	0.4	-3000	12.0	20.0	-200	-40.0	55
DATA5	2.4	0.4	-3000	12.0	20.0	-200	-40.0	55
DATA6	2.4	0.4	-3000	12.0	20.0	-200	-40.0	55
DATA7	2.4	0.4	-3000	12.0	20.0	-200	-40.0	55
-INT	2.4	0.4	-1000	12.0	NA	NA	-40.0	20
-PAUSE	2.4	0.4	-1000	12.0	NA	NA	-40.0	25
-DTACK	2.4	0.5	-2000	8.0	NA	NA	-30.0	20
PU	2.4	NA	-1000	NA	NA	NA	NA	NA

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6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The module function is printed herein in the appendix.

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6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
READ	2	I	Read/Write. This signal is high when the microprocessor is reading information from the SQY. When low, the microprocessor is in a write cycle.
-CS	3	I	Chip Select. This active low input enables read and write operations on the data bus. Interrupt acknowledge responses are not conditioned by -CS.
-EXT1 -EXT2 -EXT3 -EXT4 -EXT5	8 59 61 60 51	I I I I I	External Response-In-Progress (RIP). These input signals are used to determine if another SQY is currently responding to an interrupt.
-BES	52	I	Bus Enable Strobe. This active low input is used to enable the output data buffer allowing data transfer to and from the SQY.
-IACK	53	I	Interrupt Acknowledge. The active low Interrupt Acknowledge line indicates that the external system is asking for interrupt response information. Depending on the programmed state of SQY, it will accept 1, 2, 3 or 4 -IACK pulses; one response byte is transferred per pulse. The first -IACK pulse causes selection of the highest priority unmasked pending interrupt request and generates a -DTACK output signal.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-DS-R	54	I	Read Data Strobe. This active low signal is used to enable the read cycle to occur and -IACK pulses to be recognized. -DS-R signals the SQY when data should be placed on the data bus.
-IREQ0	66	I	Interrupt Request. The Interrupt Request signals are used by external devices to indicate that service by the microprocessor is desired. -IREQ inputs are accepted asynchronously and they may be programmed for either a high-to-low or low-to-high edge transition.
-IREQ1	68	I	
-IREQ2	70	I	
-IREQ3	21	I	
-IREQ4	24	I	
-IREQ5	23	I	
-IREQ6	71	I	
-IREQ7	16	I	
-IREQ8	93	I	
-IREQ9	96	I	
-IREQ10	97	I	
-IREQ11	48	I	
-IREQ12	100	I	
-IREQ13	50	I	
-IREQ14	98	I	
-IREQ15	95	I	
A1	32	I	Address Bit 1. This signal selects source and destination locations for data bus read and write operations. Data read or write transfers are made to or from preselected internal registers or memory locations. Control write operations load the command register and control read operations output the status register.
-DS-W	80	I	Write. This input is low when the microprocessor is in a write cycle, and allows data to be transferred from the data bus to the SQY.
PU	11	O	General purpose pull-up source of logic "1".

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-DTACK	9	O	External Data Transfer Acknowledge. This output signal becomes active (low) when RIP is generated internally by one of the AM9519As or when an external RIP (-EXT1 to -EXT5) is detected as being low.
-INT	82	O	Interrupt. This active low interrupt output signal indicates that at least one unmasked interrupt request is pending.
-PAUSE	86	O	Pause. The active low pause signal is used to coordinate interrupt responses with data bus and control timing. -PAUSE goes low when the first -IACK is received and remains low until -DTACK goes low. The external system can use -PAUSE to stretch the acknowledge cycle and allow the control timing to automatically adjust to the actual priority resolution delays in the interrupt system. Second, third, and fourth response bytes do not cause -PAUSE to go low.
DATA0 DATA1 DATA2 DATA3 DATA4 DATA5 DATA6 DATA7	79 29 26 78 28 76 27 30	I/O I/O I/O I/O I/O I/O I/O I/O	Data Bus. The eight bidirectional data bus signals are used to transfer information between the SQY and the system data bus. The direction of transfer is controlled by the -IACK and READ input signals. Programming and control information is written into the SQY; status and response data are output by the SQY.
A2	56	I	Address Bit 2. This input is used to select one of the AM9519As and insures that only one chip select is active at any time. A2=0 selects the AM9519A unit 1 and A2=1 selects AM9519A unit 2.

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 7.

(j) Vol test - table XI, figure 7.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figures 7 and 8.

(o) Dynamic test - tables XVI and XVII, figures 7, 9 and 10.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figures 8 and 11.

(r) Output crosstalk tests - tables XX and XXI, figures 7, 12, 13 and 14.

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(s) Cin test - table XXII.

(t) Cout test - table XXIII.

(u) Threshold test - table XXIV, figure 8.

(v) Disable time test - table XXV, figures 7, 15 and 16.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT, AV
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0050)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	All pins	Frame	10	INF	MOhm	1-7
A2	Isolation, used to unused pin	All used pins	All unused pins	10	INF	MOhm	1-7
A3	Isolation, unused to unused pin	99	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77,81,83, 84,85,87, 88,89,90, 91,92,94	10	INF	MOhm	1-7
A4	Isolation, unused to unused pin	94	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77,81,83, 84,85,87, 88,89,90, 91,92	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A5	Isolation, unused to unused pin	92	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77,81,83, 84,85,87, 88,89,90, 91	10	INF	MOhm	1-7
A6	Isolation, unused to unused pin	91	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77,81,83, 84,85,87, 88,89,90,	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A7	Isolation, unused to unused pin	90	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77,81,83, 84,85,87, 88,89	10	INF	MOhm	1-7
A8	Isolation, unused to unused pin	89	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77,81,83, 84,85,87, 88	10	INF	MOhm	1-7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:56Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A9	Isolation, unused to unused pin	88	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77,81,83, 84,85,87	10	INF	MOhm	1-7
A10	Isolation, unused to unused pin	87	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77,81,83, 84,85	10	INF	MOhm	1-7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:56Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A11	Isolation, unused to unused pin	85	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77,81,83, 84	10	INF	MOhm	1-7
A12	Isolation, unused to unused pin	84	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77,81,83	10	INF	MOhm	1-7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:56Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A13	Isolation, unused to unused pin	83	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77,81	10	INF	MOhm	1-7
A14	Isolation, unused to unused pin	81	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75, 77	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A15	Isolation, unused to unused pin	77	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74,75	10	INF	MOhm	1-7
A16	Isolation, unused to unused pin	75	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73,74	10	INF	MOhm	1-7
A17	Isolation, unused to unused pin	74	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72, 73	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A18	Isolation, unused to unused pin	73	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69,72	10	INF	MOhm	1-7
A19	Isolation, unused to unused pin	72	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67,69	10	INF	MOhm	1-7
A20	Isolation, unused to unused pin	69	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65, 67	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A21	Isolation, unused to unused pin	67	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64,65	10	INF	MOhm	1-7
A22	Isolation, unused to unused pin	65	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63,64	10	INF	MOhm	1-7
A23	Isolation, unused to unused pin	64	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62, 63	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A24	Isolation, unused to unused pin	63	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58,62	10	INF	MOhm	1-7
A25	Isolation, unused to unused pin	62	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57,58	10	INF	MOhm	1-7
A26	Isolation, unused to unused pin	58	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55, 57	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A27	Isolation, unused to unused pin	57	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49,55	10	INF	MOhm	1-7
A28	Isolation, unused to unused pin	55	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47,49	10	INF	MOhm	1-7
A29	Isolation, unused to unused pin	49	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46, 47	10	INF	MOhm	1-7
A30	Isolation, unused to unused pin	47	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45,46	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A31	Isolation, unused to unused pin	46	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44,45	10	INF	MOhm	1-7
A32	Isolation, unused to unused pin	45	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43, 44	10	INF	MOhm	1-7
A33	Isolation, unused to unused pin	44	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42,43	10	INF	MOhm	1-7
A34	Isolation, unused to unused pin	43	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41,42	10	INF	MOhm	1-7
A35	Isolation, unused to unused pin	42	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39, 41	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A36	Isolation, unused to unused pin	41	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38,39	10	INF	MOhm	1-7
A37	Isolation, unused to unused pin	39	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37,38	10	INF	MOhm	1-7
A38	Isolation, unused to unused pin	38	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36, 37	10	INF	MOhm	1-7
A39	Isolation, unused to unused pin	37	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35,36	10	INF	MOhm	1-7
A40	Isolation, unused to unused pin	36	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34,35	10	INF	MOhm	1-7
A41	Isolation, unused to unused pin	35	4,5,6,7, 12,13,14, 17,18,19, 20,22,33, 34	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A42	Isolation, unused to unused pin	34	4,5,6,7, 12,13,14, 17,18,19, 20,22,33	10	INF	MOhm	1-7
A43	Isolation, unused to unused pin	33	4,5,6,7, 12,13,14, 17,18,19, 20,22	10	INF	MOhm	1-7
A44	Isolation, unused to unused pin	22	4,5,6,7, 12,13,14, 17,18,19, 20	10	INF	MOhm	1-7
A45	Isolation, unused to unused pin	20	4,5,6,7, 12,13,14, 17,18,19	10	INF	MOhm	1-7
A46	Isolation, unused to unused pin	19	4,5,6,7, 12,13,14, 17,18	10	INF	MOhm	1-7
A47	Isolation, unused to unused pin	18	4,5,6,7, 12,13,14, 17	10	INF	MOhm	1-7
A48	Isolation, unused to unused pin	17	4,5,6,7, 12,13,14	10	INF	MOhm	1-7
A49	Isolation, unused to unused pin	14	4,5,6,7, 12,13	10	INF	MOhm	1-7
A50	Isolation, unused to unused pin	13	4,5,6,7, 12	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A51	Isolation, unused to unused pin	12	4,5,6,7	10	INF	MOhm	1-7
A52	Isolation, unused to unused pin	7	4,5,6	10	INF	MOhm	1-7
A53	Isolation, unused to unused pin	6	4,5	10	INF	MOhm	1-7
A54	Isolation, unused to unused pin	5	4	10	INF	MOhm	1-7

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V 30 V	± 10 percent ± 10 percent	A1 A2-A54
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A1-A54 A1-A54

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity,Vcc	1	31	0.0	1.0	Ohm	1-4
B2	Continuity,GND	10	15				
B3	Continuity,GND	15	25				
B4	Continuity,GND	25	40	0.0	1.0	Ohm	1-4
B5	Continuity,Vcc	1	31	0.0	0.1	Ohm	1-4
B6	Continuity,GND	10	15				
B7	Continuity,GND	15	25				
B8	Continuity,GND	25	40	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B4 B5-B8

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31	10,15 25,40	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	1.356 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	READ	2	5.5	0.0	1.1	mA	1-3
D2	-CS	3	5.5	0.0	1.0	mA	1-3
D3	-EXT1	8					
D4	-EXT2	59					
D5	-EXT3	61					
D6	-EXT4	60					
D7	-EXT5	51					
D8	-BES	52					
D9	-IACK	53					
D10	-DS-R	54					
D11	A2	56	5.5	0.0	1.0	mA	1-3
D12	-IREQ0	66	5.5	0.0	0.13	mA	1-3
D13	-IREQ1	68					
D14	-IREQ2	70					
D15	-IREQ3	21					
D16	-IREQ4	24					
D17	-IREQ5	23					
D18	-IREQ6	71					
D19	-IREQ7	16					
D20	-IREQ8	93	5.5	0.0	0.13	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	-IREQ9	96	5.5	0.0	0.13	mA	1-3
D22	-IREQ10	97					
D23	-IREQ11	48					
D24	-IREQ12	100					
D25	-IREQ13	50					
D26	-IREQ14	98					
D27	-IREQ15	95	5.5	0.0	0.13	mA	1-3
D28	A1	32	5.5	0.0	0.1	mA	1-3
D29	-DS-W	80					
D30	DATA0	79					
D31	DATA1	29					
D32	DATA2	26					
D33	DATA3	78					
D34	DATA4	28					
D35	DATA5	76					
D36	DATA6	27					
D37	DATA7	30	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V 5.5 V 5.5 V	± 55 mV + 0 mV, -55 mV ± 55 mV	D1-D11 D12-D27 D28-D37
Power supply current limit	1.13 A	NA	D1-D37
Forcing voltage, (Vin)	5.5 V 5.5 V 5.5 V	± 55 mV - 0 mV, +55 mV ± 55 mV	D1-D11 D12-D27 D28-D37
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	D1-D37 D1-D37
Input current measurement	NA NA NA NA	± 55.0 uA ± 50.0 uA ± 6.5 uA ± 5.0 uA	D1 D2-D11 D12-D27 D28-D37

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	READ	2	-18.0	-1.5	-0.4	V	1-3
E2	-CS	3					
E3	-EXT1	8					
E4	-EXT2	59					
E5	-EXT3	61					
E6	-EXT4	60					
E7	-EXT5	51					
E8	-BES	52					
E9	-IACK	53					
E10	-DS-R	54					
E11	A2	56					
E12	-IREQ0	66					
E13	-IREQ1	68					
E14	-IREQ2	70					
E15	-IREQ3	21					
E16	-IREQ4	24					
E17	-IREQ5	23					
E18	-IREQ6	71					
E19	-IREQ7	16					
E20	-IREQ8	93	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	-IREQ9	96	-18.0	-1.5	-0.4	V	1-3
E22	-IREQ10	97					
E23	-IREQ11	48					
E24	-IREQ12	100					
E25	-IREQ13	50					
E26	-IREQ14	98					
E27	-IREQ15	95					
E28	A1	32					
E29	-DS-W	80					
E30	DATA0	79					
E31	DATA1	29					
E32	DATA2	26					
E33	DATA3	78					
E34	DATA4	28					
E35	DATA5	76					
E36	DATA6	27					
E37	DATA7	30	-18.0	-1.5	-0.4	V	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:56Z
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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E37
Power supply current limit	1.13 A	NA	E1-E37
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E37
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E37 E1-E37
Input voltage measurement	NA	± 55 mV	E1-E37

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	-DTACK	9	0.0	-130	-30	mA	1-4
F2	-INT	82	0.0	-225	-40	mA	1-4
F3	-PAUSE	86					
F4	DATA0	79					
F5	DATA1	29					
F6	DATA2	26					
F7	DATA3	78					
F8	DATA4	28					
F9	DATA5	76					
F10	DATA6	27					
F11	DATA7	30	0.0	-225	-40	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F11
Power supply current limit	1.13 A	NA	F1-F11
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F11
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F11 F1-F11
Output current measurement	 NA NA	 ± 5.0 mA ± 9.25 mA	 F1 F2-F11

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	READ	2	2.4	0	45	uA	1-3
G2	-CS	3	2.4	0	25	uA	1-3
G3	-EXT1	8					
G4	-EXT2	59					
G5	-EXT3	61					
G6	-EXT4	60					
G7	-EXT5	51					
G8	-BES	52					
G9	-IACK	53					
G10	-DS-R	54					
G11	A2	56	2.4	0	25	uA	1-3
G12	-IREQ0	66	2.7	-887	-783	uA	1-3
G13	-IREQ1	68					
G14	-IREQ2	70					
G15	-IREQ3	21					
G16	-IREQ4	24					
G17	-IREQ5	23					
G18	-IREQ6	71					
G19	-IREQ7	16					
G20	-IREQ8	93	2.7	-887	-783	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	-IREQ9	96	2.7	-887	-783	uA	1-3
G22	-IREQ10	97					
G23	-IREQ11	48					
G24	-IREQ12	100					
G25	-IREQ13	50					
G26	-IREQ14	98					
G27	-IREQ15	95	2.7	-887	-783	uA	1-3
G28	A1	32	2.7	0	20	uA	1-3
G29	-DS-W	80	2.7	0	20	uA	1-3
G30	DATA0	79	2.7	-20	20	uA	1-3
G31	DATA1	29					
G32	DATA2	26					
G33	DATA3	78					
G34	DATA4	28					
G35	DATA5	76					
G36	DATA6	27					
G37	DATA7	30	2.7	-20	20	uA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G37
Power supply current limit	1.13 A	NA	G1-G37
Forcing voltage, (Vih)	2.4 V 2.7 V	± 120 mV ± 135 mV	G1-G11 G12-G37
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G37 G1-G37
Input current measurement	NA NA NA NA NA	± 2.25 uA ± 1.25 uA ± 5.2 uA ± 1.0 uA ± 2.0 uA	G1 G2-G11 G12-G27 G28,G29 G30-G37

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	READ	2	0.4	-450	-5	uA	1-3
H2	-CS	3	0.4	-250	0	uA	1-3
H3	-EXT1	8					
H4	-EXT2	59					
H5	-EXT3	61					
H6	-EXT4	60					
H7	-EXT5	51					
H8	-BES	52					
H9	-IACK	53					
H10	-DS-R	54					
H11	A2	56	0.4	-250	0	uA	1-3
H12	-IREQ0	66	0.4	-1817	-1468	uA	1-3
H13	-IREQ1	68					
H14	-IREQ2	70					
H15	-IREQ3	21					
H16	-IREQ4	24					
H17	-IREQ5	23					
H18	-IREQ6	71					
H19	-IREQ7	16					
H20	-IREQ8	93	0.4	-1817	-1468	uA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	-IREQ9	96	0.4	-1817	-1468	uA	1-3
H22	-IREQ10	97					
H23	-IREQ11	48					
H24	-IREQ12	100					
H25	-IREQ13	50					
H26	-IREQ14	98					
H27	-IREQ15	95	0.4	-1817	-1468	uA	1-3
H28	A1	32	0.4	-400	0	uA	1-3
H29	-DS-W	80	0.4	-400	0	uA	1-3
H30	DATA0	79	0.4	-200	-5	uA	1-3
H31	DATA1	29					
H32	DATA2	26					
H33	DATA3	78					
H34	DATA4	28					
H35	DATA5	76					
H36	DATA6	27					
H37	DATA7	30	0.4	-200	-5	uA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H37
Power supply current limit	1.13 A	NA	H1-H37
Forcing voltage, (Vil)	0.4 V	± 20 mV	H1-H37
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H37
low, (VL)	0.0 V	± 100 mV	H1-H37
Input current measurement	NA	± 22.3 uA	H1
	NA	± 12.5 uA	H2-H11
	NA	± 17.5 uA	H12-H27
	NA	± 20.0 uA	H28,H29
	NA	± 9.75 uA	H30-H37

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	PU	11	-1.0	2.4	2.6	V	1-5
J2	-DTACK	9	-2.0	2.4	4.5	V	1-5
J3	-INT	82	-1.0	2.4	4.5	V	1-5
J4	-PAUSE	86	-1.0	2.4	4.5	V	1-5
J5	DATA0	79	-3.0	2.4	4.5	V	1-5
J6	DATA1	29					
J7	DATA2	26					
J8	DATA3	78					
J9	DATA4	28					
J10	DATA5	76					
J11	DATA6	27					
J12	DATA7	30	-3.0	2.4	4.5	V	1-5

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J12
Power supply current limit	1.13 A	NA	J1-J12
Forcing current, (Ioh)	-1.0 -2.0 -1.0 -3.0	± 10 uA ± 20 uA ± 10 uA ± 30 uA	J1 J2 J3,J4 J5-J12
Pattern voltage high, (VH) low, (VL)	See note 4 See note 4		J1-J12 J1-J12
Output voltage measurement	NA NA	± 10 mV ± 105 mV	J1 J2-J12

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

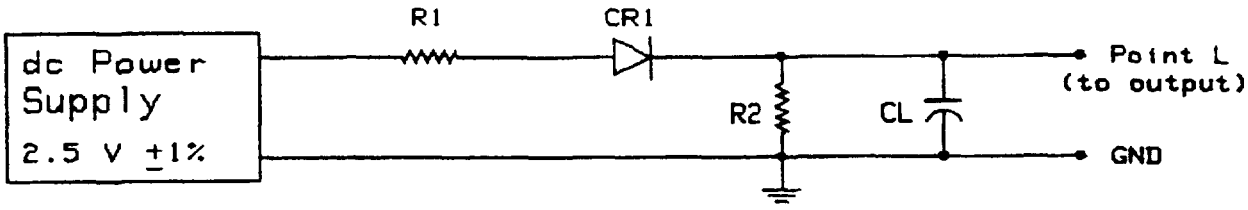
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 7.

4. The low and high input threshold voltage for each pin is as specified in 6.3.1.

5. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) CL = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) CL = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) CL = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) CL not required for any other tests.

FIGURE 7. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are listed below.

4. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
9	1200	8.0	0.5
30,27,79,29,26, 78,28,76	800	12.0	0.4
82,86	2400	12.0	0.4
11	2400	NA	NA

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K1	-DTACK	9	8.0	0.0	0.5	V	1-5
K2	-INT	82	12.0	0.0	0.4	V	1-5
K3	-PAUSE	86					
K4	DATA0	79					
K5	DATA1	29					
K6	DATA2	26					
K7	DATA3	78					
K8	DATA4	28					
K9	DATA5	76					
K10	DATA6	27					
K11	DATA7	30	12.0	0.0	0.4	V	1-5

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K11
Power supply current limit	1.13 A	NA	K1-K11
Forcing current, (Iol)	8.0 V 12.0 V	± 80 uA ± 120 uA	K1 K2-K11
Pattern voltage high, (VH) low, (VL)	 See note 4 See note 4		 K1-K11 K1-K11
Output voltage measurement	NA NA	± 25 mV ± 20 mV	K1 K2-K11

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 7.

4. The input low and high threshold voltage for each pin is as specified in 6.3.1.

5. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	DATA0	79	2.7	-20	20	uA	1-3
L2	DATA1	29					
L3	DATA2	26					
L4	DATA3	78					
L5	DATA4	28					
L6	DATA5	76					
L7	DATA6	27					
L8	DATA7	30	2.7	-20	20	uA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L8
Power supply current limit	1.13 A	NA	L1-L8
Forcing voltage, (Voh)	2.7 V	± 135 mV	L1-L8

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 L1-L8 L1-L8
Output current measurement	NA	± 2 uA	L1-L8

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	DATA0	79	0.4	-200	20	uA	1-3
M2	DATA1	29					
M3	DATA2	26					
M4	DATA3	78					
M5	DATA4	28					
M6	DATA5	76					
M7	DATA6	27					
M8	DATA7	30	0.4	-200	20	uA	1-3

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M8
Power supply current limit	1.13 A	NA	M1-M8
Forcing voltage, (Vol)	0.4 V	± 20 mV	M1-M8

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 M1-M8 M1-M8
Output current measurement	NA	± 11 uA	M1-M8

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31	70	754	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.13 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 34.2 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-3
P2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.13 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 150 pF (see figure 7). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

3. Abbreviations used are defined in 6.1.

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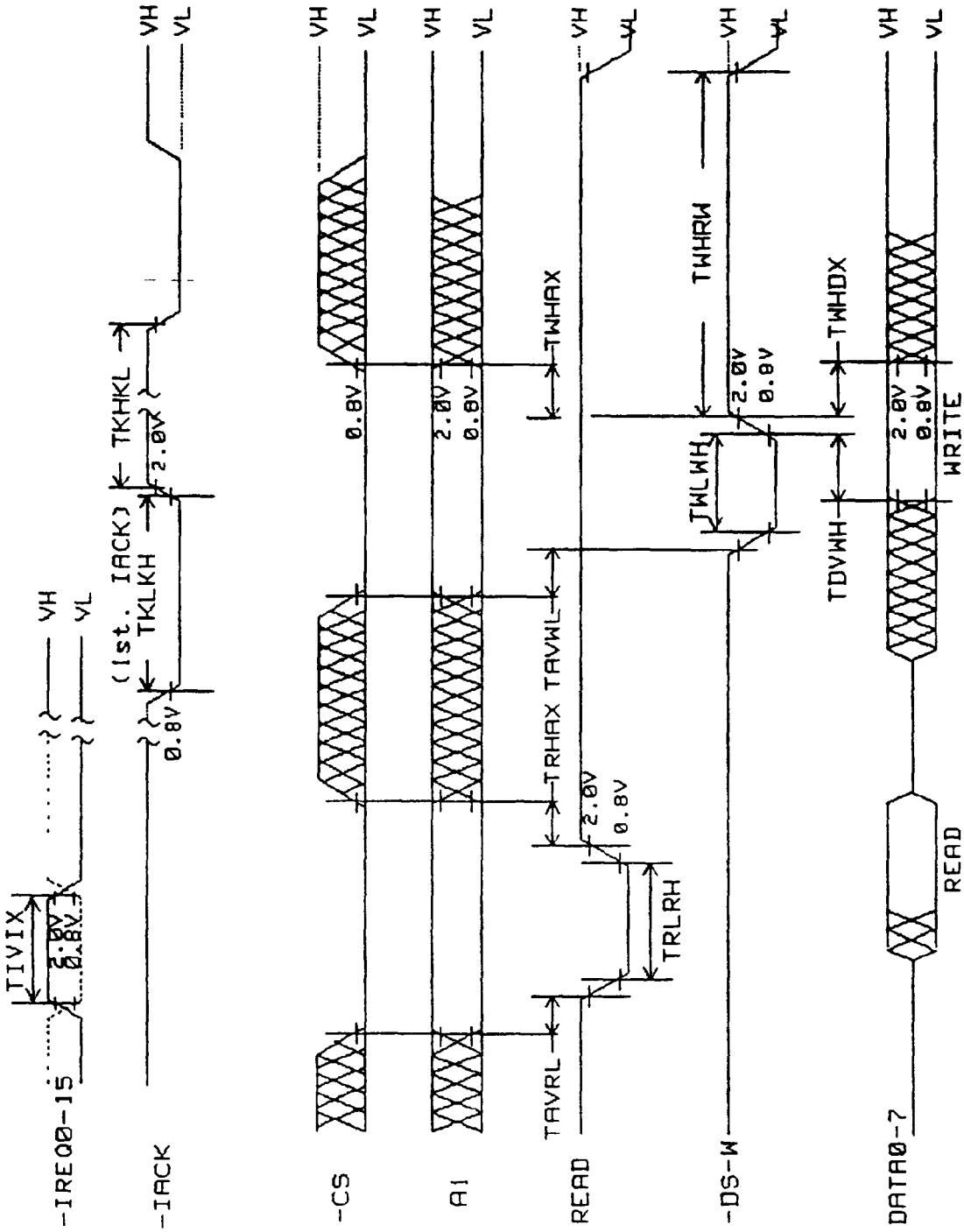


FIGURE 8. Input timing parameters for table XV, XIX and XXIV.

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NOTES:

1. Input parameters for figure 8.

   TIVIX = 271 nS. min.
   TKLKH = 1020 nS. min.
   TKHKL = 545 nS. min.
   TAVRL = 45 nS. min.
   TAVWL = 43 nS. min.
   TDVWH = 270 nS. min.
   TRHAX = 45 nS. min.
   TRLRH = 345 nS. min.
   TWHAX = 45 nS. min.
   TWHDX = 45 nS. min.
   TWHRW = 620 nS. min.
   TWLWH = 322 nS. min.

2. Rise times (as measured between the 10 percent and 90 percent points) shall be no more than 12 nS.

3. Fall times (as measured between the 10 percent and 90 percent points) shall be no more than 10 nS.

4. VH and VL are as specified in the test tolerance table.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes	Figs
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-9	9,10
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-9	9,10
Q3	Dynamic test, qualification maximum pattern rate minimum power supply	1-9	9,10
Q4	Dynamic test, qualification maximum pattern rate maximum power supply	1-9	9,10

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.13 A	NA	Q1,Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105 mV	Q1,Q3
high, (Voh)	NA	± 155 mV	Q2,Q4
low, (Vol)	NA	± 25 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $2.4V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

3. The load circuit for this test is specified on figure 7.

4. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

5. During the first -IACK pulse, -PAUSE will be low long enough to allow for priority resolution and will not go high until after -DTACK goes low.

6. This parameter applies only to second, third, and fourth -IACK pulses while -DTACK is low. During the first -IACK pulse, DATA 0-7 will be valid following the falling edge of -DTACK.

7. -DTACK is pulled low to indicate that an interrupt request has been selected. -DTACK remains low until after the rising edge of the -IACK pulse that transfers the last response byte for the selected interrupt request.

8. -CS must be high for at least 145 nS prior to -IACK going low.

9. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
1	1-16	-IREQ0--IREQ15 pulse duration	271	266	271	-	-	-	nS	10a
2	17-32	-IREQ0--IREQ15 active to -INT	104	104	104	858	845	858	nS	10a
3	33-34	-IACK low to -DTACK low	75	75	75	745	745	745	nS	10a
4	35-36	-IACK high to -DTACK high	0	0	0	545	545	545	nS	10a
5	37-38	-DTACK low to -PAUSE high	32	32	32	406	401	406	nS	10a
6	39-70	1st -IACK low to DATA0-DATA7 valid	77	77	77	728	724	728	nS	10a
7	71-102	-DTACK low to DATA0-DATA7 valid	-43	-43	-43	83	79	83	nS	10a
8	103-104	-IACK high to -IACK low (-IACK recovery)	545	545	545	-	-	-	nS	10a
9	105-106	-IACK low to -IACK high (1st -IACK)	1020	1020	1020	-	-	-	nS	10a
11	107-138	-IACK low to DATA0-DATA7 valid	27	27	27	378	374	378	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

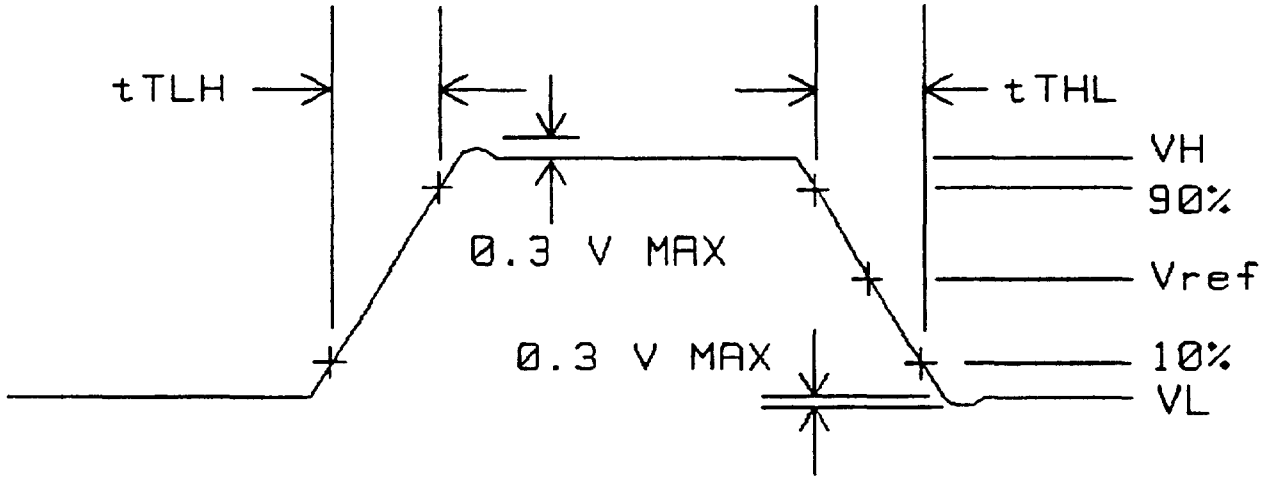
Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
13	141-142	READ pulse duration	345	345	345	-	-	-	nS	10b
14	143-144	-DS-W (write pulse duration)	322	316	322	-	-	-	nS	10b
16	145-176	READ high to DATA0-DATA7 valid	-	-	-	378	374	378	nS	10b
18	177-208	DATA0-DATA7 input valid to -DS-W high	270	266	270	-	-	-	nS	10b
19	209-240	-DS-W high to DATA0-DATA7 in unknown	45	45	45	-	-	-	nS	10b
20	241-242	A1 and -CS low to READ low	45	45	45	-	-	-	nS	10b
21	243-244	A1 and -CS low to -DS-W low	43	43	43	-	-	-	nS	10b
22	245-246	READ high to A1 and -CS don't care	45	45	45	-	-	-	nS	10b
23	247-248	-DS-W high to A1 to -CS don't care	45	41	45	-	-	-	nS	10b
6	255-286	1st -IACK low to DATA0-DATA7 valid	77	77	77	742	738	742	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
7	287-318	-DTACK low to DATA0-DATA7 valid	-43	-43	-43	97	93	97	nS	10a
11	319-350	-IACK low to DATA0-DATA7 valid	27	27	27	392	388	392	nS	10a
16	351-382	READ low DATA0-DATA7 valid	-	-	-	392	388	392	nS	10b
18	383-414	DATA0-DATA7 input valid to -DS-W high	270	266	270	-	-	-	nS	10b
19	415-446	-DS-W high to DATA0-DATA7 unknown	45	45	45	-	-	-	nS	10b
24	249-250	-DS-W low to -DS-W high (write recovery)	620	616	620	-	-	-	nS	10b
24	251-252	-DS-W high to READ high (write recovery)	620	616	620	-	-	-	nS	10b
25	253-254	-IACK low to -PAUSE low	-	-	-	256	252	256	nS	10a

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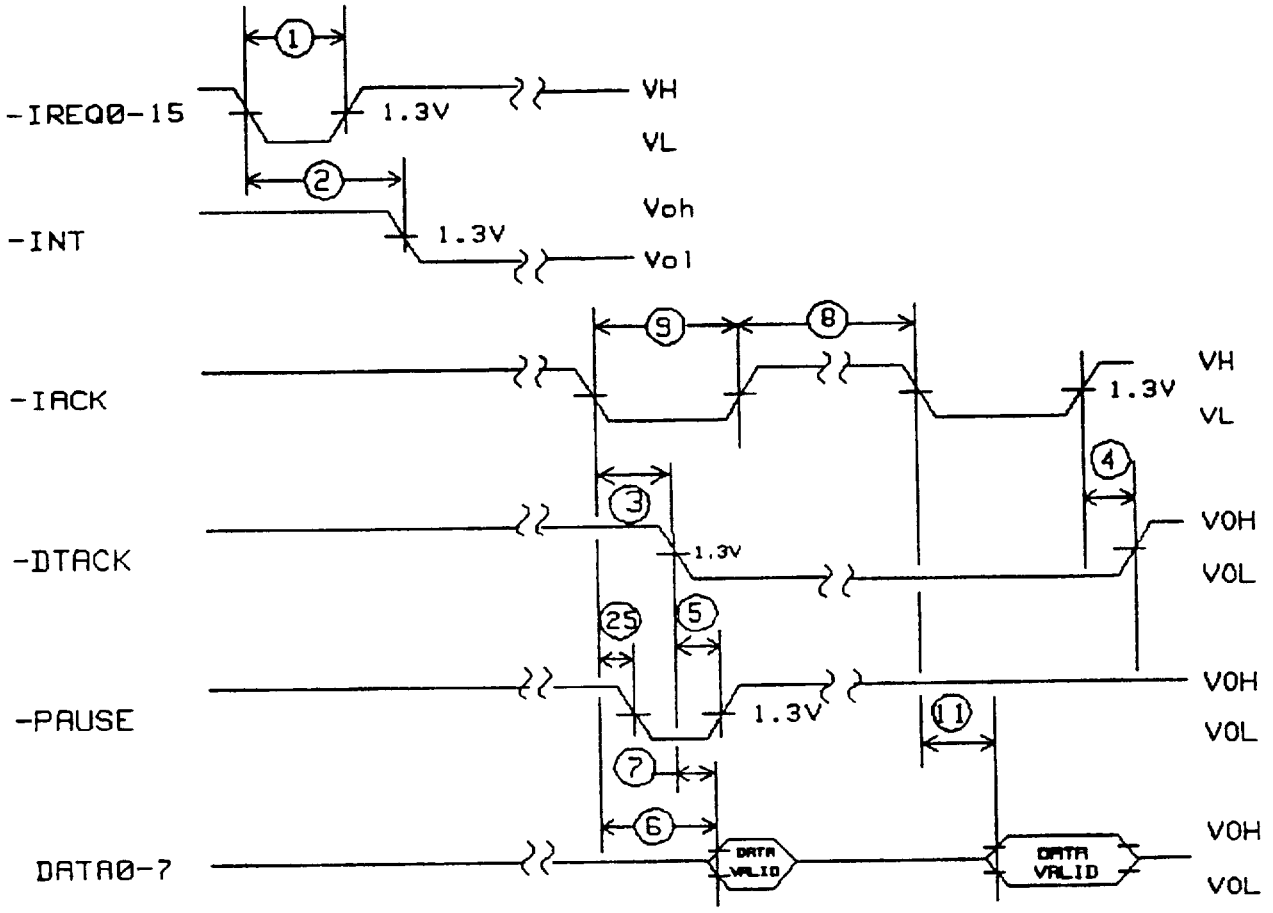
NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.5 V.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
66,68,70,21,24,23,71,16,80, 32,2,52,3,53,54,56,59,8,61, 60,51,93,96,97,48,100,50,98, 95	15 ± 1.5 nS	6 ± 1.5 nS
79,29,26,78,28,76,27,30	15 ± 1.5 nS	6 ± 1.5 nS

FIGURE 9. Input pulse parameters for tables XVI, XVII, and XXV.

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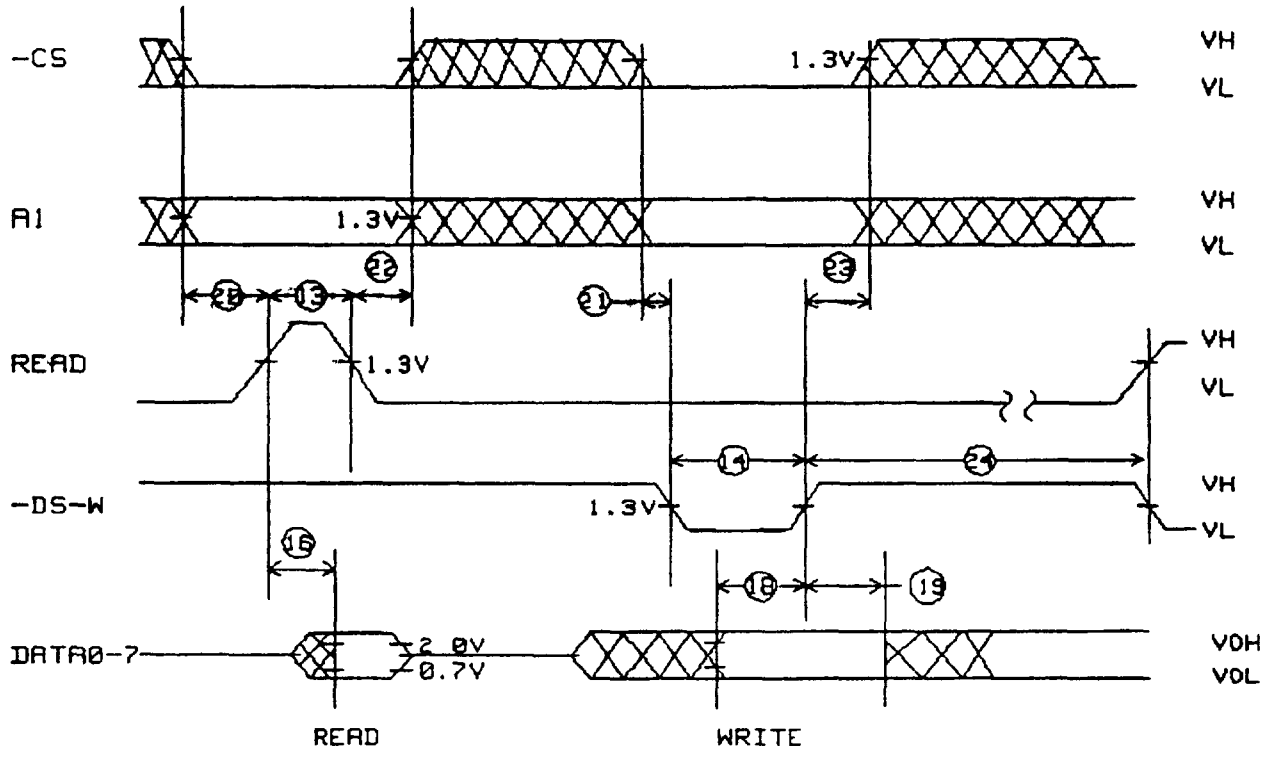


NOTES:

1. Valid output high = 2.0 V, valid output low = 0.7 V.
2. VH and VL are as specified in the test tolerance table.

FIGURE 10a. Dynamic waveforms for tables XVI and XVII.

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NOTE:

1. VH and VL are as specified in the test tolerance table.

FIGURE 10b. Dynamic waveforms for tables XVI and XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	3.7	6.9	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 160 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
S	PSTA	1,31	0	400	mV	1-3	11

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	1.13 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX. The module inputs shall meet the timing requirements shown on figure 8.

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NOTES: - Continued.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D07, inspect for the most negative spike as shown on figure 11 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

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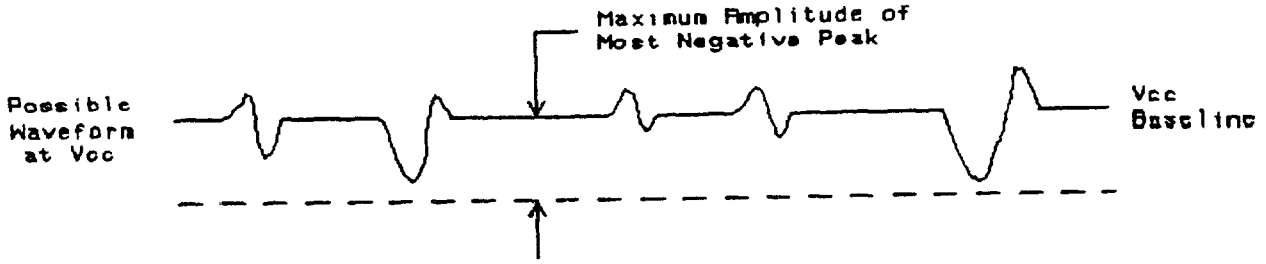


FIGURE 11. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Output crosstalk, high level, test.

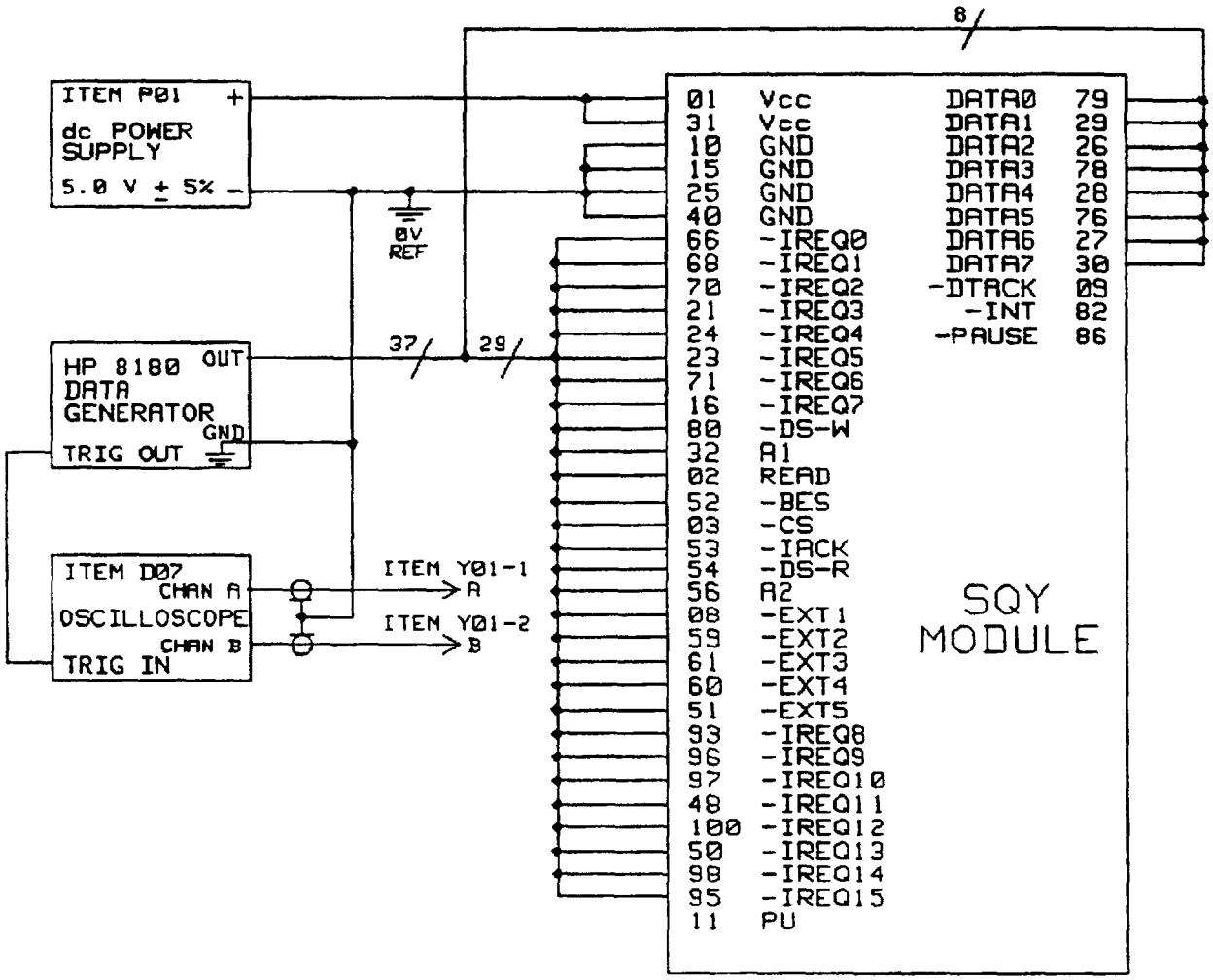
Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	-INT	82	0	400	mV	1-4	12-14
T2	-PAUSE	86					
T3	DATA0	79					
T4	DATA1	29					
T5	DATA2	26					
T6	DATA3	78					
T7	DATA4	28					
T8	DATA5	76					
T9	DATA6	27					
T10	DATA7	30	0	400	mV	1-4	12-14
T11	-DTACK	9	0	500	mV	1-4	12-14

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TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
U1	-INT	82	0	400	mV	1-4	12-14
U2	-PAUSE	86					
U3	DATA0	79					
U4	DATA1	29					
U5	DATA2	26					
U6	DATA3	78					
U7	DATA4	28					
U8	DATA5	76					
U9	DATA6	27					
U10	DATA7	30	0	400	mV	1-4	12-14
U11	-DTACK	9	0	500	mV	1-4	12-14

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NOTE:

1. Outputs shall be loaded as specified on figure 7.

FIGURE 12. Output crosstalk, high and low level, test circuit for tables XX and XXI.

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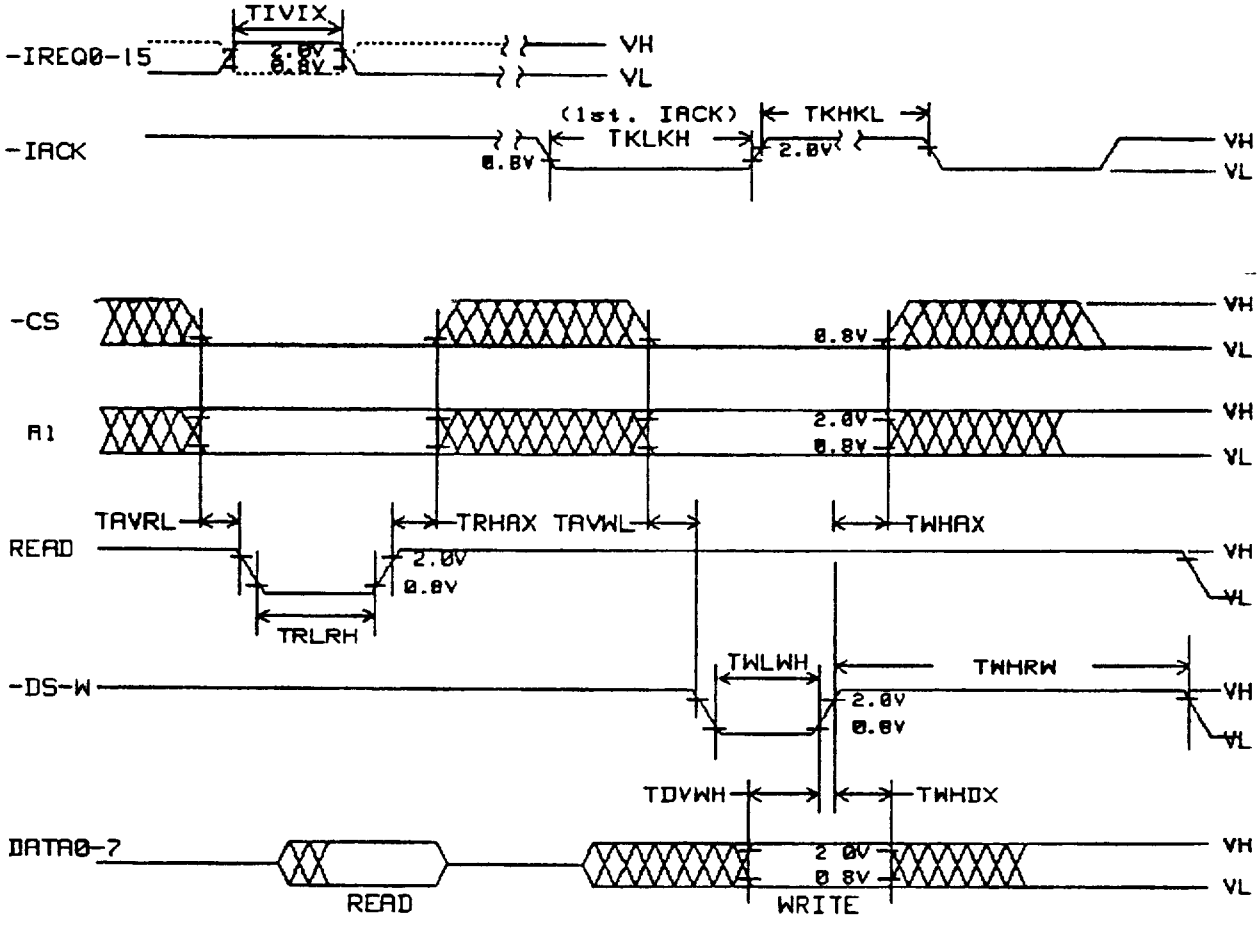


FIGURE 13. Module input timing relationship for tables XX and XXI.

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NOTES:

1. Input parameters:

TIVIX = 271 nS. min.
TKLKH = 1020 nS. min.
TKHKL = 545 nS. min.
TAVRL = 45 nS. min.
TAVWL = 43 nS. min.
TDVWH = 270 nS. min.
TRHAX = 45 nS. min.
TRLRH = 345 nS. min.
TWHAX = 45 nS. min.
TWHDX = 45 nS. min.
TWHRW = 620 nS. min.
TWLWH = 322 nS. min.

2. Rise times (as measured between the 10 percent and 90 percent points) shall be no more than 12 nS.

3. Fall times (as measured between the 10 percent and 90 percent points) shall be no more than 10 nS.

4. VH and VL are as specified in the test tolerance table.

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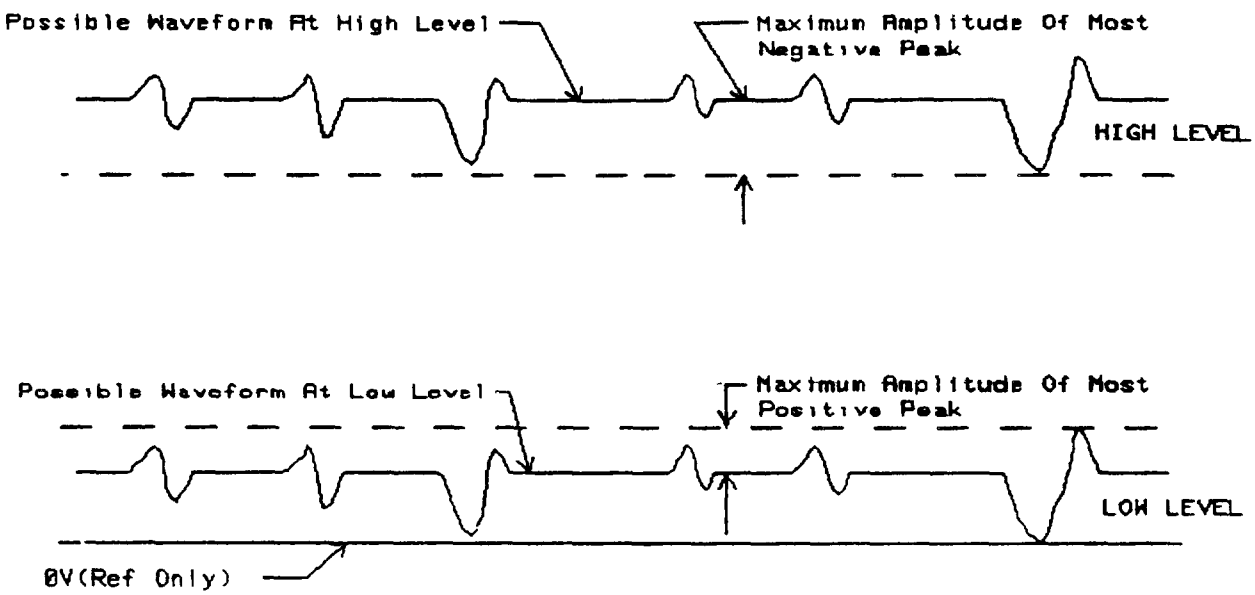


FIGURE 14. Output crosstalk test circuit waveforms, high and low level, for tables XX and XXI.

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TABLE XXII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
V1	READ	2	4	90	pF	1-3
V2	-CS	3	2	15	pF	1-3
V3	-EXT1	8	2	15	pF	1-3
V4	-EXT2	59	2	20	pF	1-3
V5	-EXT3	61	2	30	pF	1-3
V6	-EXT4	60	2	25	pF	1-3
V7	-EXT5	51	2	25	pF	1-3
V8	-BES	52	2	20	pF	1-3
V9	-IACK	53	2	15	pF	1-3
V10	-DS-R	54				
V11	A2	56	2	15	pF	1-3
V12	-IREQ7	16	2	40	pF	1-3
V13	-IREQ3	21				
V14	-IREQ5	23				
V15	-IREQ4	24				
V16	-IREQ11	48				
V17	-IREQ13	50				
V18	-IREQ0	66				
V19	-IREQ1	68				
V20	-IREQ2	70	2	40	pF	1-3

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TABLE XXII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
V21	-IREQ6	71	2	40	pF	1-3
V22	-IREQ8	93				
V23	-IREQ15	95				
V24	-IREQ9	96				
V25	-IREQ10	97				
V26	-IREQ14	98				
V27	-IREQ12	100	2	40	pF	1-3
V28	A1	32	2	15	pF	1-3
V29	-DS-W	80	2	15	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	V1-V29
Capacitance meter voltage	15 mV	± 1.5 mV	V1-V29

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 4.3 pF	V1
	NA	± 0.65 pF	V2,V3
	NA	± 0.9 pF	V4
	NA	± 1.4 pF	V5
	NA	± 1.15 pF	V6,V7
	NA	± 0.9 pF	V8
	NA	± 0.65 pF	V9-V11
	NA	± 1.9 pF	V12-V27
	NA	± 0.65 pF	V28,V29

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIII. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W1	-DTACK	9	2	20	pF	1-3
W2	-INT	82	2	20	pF	1-3
W3	-PAUSE	86	2	25	pF	1-3
W4	DATA2	26	2	55	pF	1-3
W5	DATA6	27				
W6	DATA4	28				
W7	DATA1	29				
W8	DATA7	30				
W9	DATA5	76				
W10	DATA3	78				
W11	DATA0	79	2	55	pF	1-3

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NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W11
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W11
Capacitance measurement	NA NA NA	± 0.9 pF ± 1.15 pF ± 2.65 pF	W1,W2 W3 W4-W11

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Threshold test.

Test cond	Parameter	Notes	Fig
X	Threshold	1-4	8

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	X
Power supply current limit	1.13 A	NA	X
Pattern voltage			
high, (VH)	See note 3		X
low, (VL)	See note 3		X
Output voltage measurement			
high, (Voh)	NA	± 105 mV	X
high, (Voh)	NA	± 100 mV	X
low, (Vol)	NA	± 25 mV	X

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results to the output low and high levels specified in the output voltage tests. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. The input low and high level threshold voltage for each input is as specified in 6.3.1.

4. Abbreviations used are defined in 6.1.

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TABLE XXV. Disable time test.

Test cond	Signal name	C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs	Notes
High to high impedance (tPHZ)							
Y1	-BES to DATA0	79	2	84	nS	15	1-5
Y2	-BES to DATA1	29					
Y3	-BES to DATA2	26					
Y4	-BES to DATA3	78					
Y5	-BES to DATA4	28					
Y6	-BES to DATA5	76					
Y7	-BES to DATA6	27					
Y8	-BES to DATA7	30					
Low to high impedance (tPLZ)							
Y9	-BES to DATA0	79					
Y10	-BES to DATA1	29					
Y11	-BES to DATA2	26					
Y12	-BES to DATA3	78					
Y13	-BES to DATA4	28					
Y14	-BES to DATA5	76					
Y15	-BES to DATA6	27					
Y16	-BES to DATA7	30	2	84	nS	15	1-5

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NOTES:

1. Disable time test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 50 mV	Y1-Y16
Power supply current limit	1.09 A	NA	Y1-Y16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	Y1-Y16 Y1-Y16
Output time measurement	NA	± 4.4 nS	Y1-Y16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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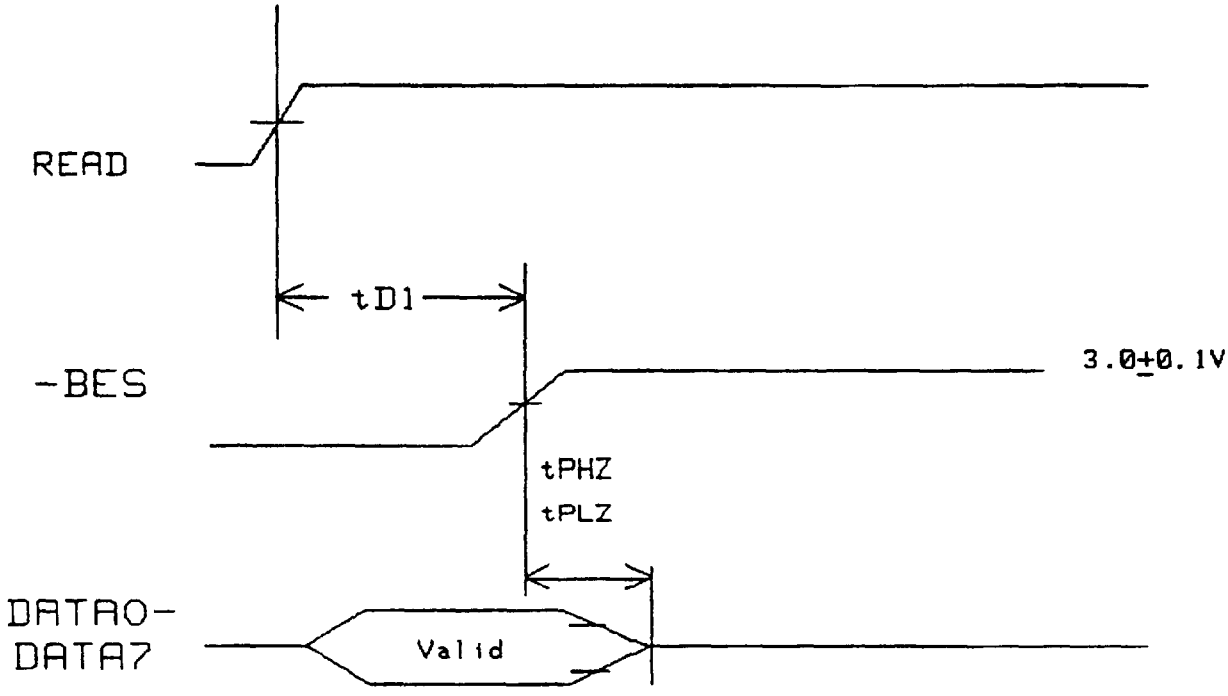
NOTES: - Continued.

3. Module inputs shall not violate the minimum input timing parameters referenced on figure 8.

4. Output pins 79,29,26,78,28,76,27 and 30 shall be loaded as specified on figure 7, except that dc supply voltage shall be 2.1 V $\pm$ 1 percent.

5. Abbreviations used are defined in 6.1.

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NOTES:

1. tTHL = tTLH as per figure 9.
2. tD1 = 500 nS. minimum.

FIGURE 15. Disable time test waveforms for table XXV.

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APPENDIX

10. SCOPE

10.1 Scope. This appendix contains information regarding the module function. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

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30. Module function. The SQY module is capable of handling up to 16 interrupt request inputs in a microprocessor based system. It is designed to prioritize these interrupt requests and provide the appropriate exception vector for the received interrupt which requests service. It will provide the appropriate exception vector for that interrupt which is the highest priority. The module design is based upon two 9519A Universal Interrupt Controllers which are cascaded together. These two devices are responsible for detecting interrupt requests, prioritizing them, generating a group interrupt to the microprocessor, and responding to an interrupt acknowledge with an exception vector.

30.1 Interrupt. The module can be programmed for fixed or rotating priority. It can also support polling as an alternative to the interrupt structure. Interrupt requests can be set in software as well as hardware. Spurious interrupt requests are ignored due to pulse-catching logic incorporated in the 9519A.

30.1.1 Programmable logic. Programmable Array Logic (PAL) is used to perform all logic functions and control timing generation on the module. The boolean expressions describing the PAL are shown in figure 1.

30.1.2 Signal buffering. All bidirectional and input signals are buffered on the module to minimize undershoot. Buffering is accomplished by using bus transceivers, low power Schottky buffers and PALS.

30.1.3 Interrupt sequence. A typical sequence of events is as follows: An -INT (interrupt output) signal is sent to the microprocessor whenever any unmasked interrupt input is active. The SQY now expects an interrupt acknowledge (-IACK) from the microprocessor. When the SQY receives -IACK from the processor, -PAUSE is brought low and the highest priority, unmasked, active interrupt request is selected. When selection is done, Response-In-Progress (-DTACK) is brought low indicating to the system that the SQY is transferring the response vector for the selected interrupt to the microprocessor. -PAUSE goes high when -DTACK goes low. -DTACK goes high after the last byte of the response memory has been transferred.

30.1.4 Reset. A reset function is accomplished by a software command (see figure 2) or automatically during power up. The resulting state of the module is that all registers are cleared and the mask register is set. Each 9519A must be reset individually.

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-RD = $\overline{\text{READ}}$ + -DS-R
-CS0 = $\overline{\text{A2}}$ + -CS
-CS1 = $\overline{\text{A2}}$ + -CS
-BUFFER EN = -BES + (-CS • -IACK)
-IACKB = -IACK + -DS-R
-DTACK = -RIP• -EXT1• -EXT2• -EXT3• -EXT4 • -EXT5

FIGURE 1. SQY PAL booleans.

COMMAND CODE								COMMAND DESCRIPTION
7	6	5	4	3	2	1	0	
0	0	0	0	0	0	0	0	Reset
0	0	0	1	0	X	X	X	Clear all IRR and all IMR bits
0	0	0	1	1	B2	B1	B0	Clear all IRR and all IMR bits specified by B2,B1,B0
0	0	1	0	0	X	X	X	Clear all IMR bits
0	0	1	0	1	B2	B1	B0	Clear IMR bit specified by B2,B1,B0
0	0	1	1	0	X	X	X	Set all IMR bits
0	0	1	1	1	B2	B1	B0	Set IMR bits specified by B2,B1,B0
0	1	0	0	0	X	X	X	Clear all IRR bits
0	1	0	0	1	B2	B1	B0	Clear IRR bits specified by B2,B1,B0
0	1	0	1	0	X	X	X	Set all IRR bits
0	1	0	1	1	B2	B1	B0	Set IRR bit specified by B2,B1,B0
0	1	1	0	X	X	X	X	Clear highest priority ISR bit
0	1	1	1	0	X	X	X	Clear all ISR bits
0	1	1	1	1	B2	B1	B0	Clear ISR bit specified by B2,B1,B0
1	0	9	M4	M3	M2	M1	M0	Load Mode register bits 0-4 with specified pattern
1	0	1	0	M6	M5	0	0	Load Mode register bits 5,6 with specified pattern
1	0	1	0	M6	M5	0	1	Load Mode register bits 5,6 and set mode bit 7
1	0	1	0	M6	M5	1	0	Load Mode register bits 5,6 and clear mode bit 7
1	0	1	1	X	X	X	X	Preselect IMR for subsequent loading from data bus
1	1	0	0	X	X	X	X	Preselect Auto Clear register for subsequent loading from data bus
1	1	1	BY1	BY0	L2	L1	L0	Load BY1, BY0 into byte count register and preselect response memory level specified by L2,L1,L0 for subsequent loading from data bus

FIGURE 2. Command summary.

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30.2 Data bus transfers. Figure 3 shows the control signal configurations for all information transfers between the SQY and the system data bus. Only one 9519A can be written to or read from at a time. The registers contained in each 9519A controller are as follows:

30.2.1 Interrupt Request Register (IRR). The 8-bit IRR is used to store pending interrupt requests. A bit in the IRR is set whenever the corresponding IREQ input goes active. Bits may also be set under program control from the microprocessor, thus permitting software generated interrupts. IRR bits may be cleared under program control. An IRR bit is automatically cleared when its interrupt is acknowledged. All IRR bits are cleared by a reset function.

30.2.2 Interrupt Service Register (ISR). The 8-bit ISR contains one bit for each IREQ input. It is used to indicate that a pending interrupt has been acknowledged and to mask all lower priority interrupts. When a bit is set by the acknowledge logic in the ISR, the corresponding IRR bit is cleared. If an acknowledged interrupt is not programmed to be automatically cleared, its ISR bit must be cleared by the microprocessor under program control when it is desired to permit interrupts from lower priority devices. When the interrupt is programmed for automatic clearing, the ISR bit is automatically reset during the acknowledge sequence. All ISR bits are cleared by a reset function.

30.2.3 Interrupt Mask Register (IMR). The 8-bit IMR is used to enable or disable the individual interrupt inputs. The IMR bits correspond to the IREQ inputs and all eight may be set or cleared in parallel under program control. In addition, individual IMR bits may be set or cleared by software. Care must be taken when disabling a channel by setting its IMR bit. If that bit is causing the GINT pin to be active, a lock-up condition can occur if the microprocessor recognizes the interrupt and then the SQY removes the request. A reset function will set all eight mask bits, disabling all requests. Only unmasked interrupt inputs can generate a GINT.

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30.2.4 Response memory. An 8x32 read/write response memory is included in the AM9519A. It is used to store up to four bytes of response information for each of the eight interrupt request inputs. All bits in the memory are programmable, allowing any desired vector, opcode, instruction or other data to be entered. THe AM9519A transfers the interrupt response information for the highest priority unmasked interrupt from the memory to the data bus when the -IACK or -DS-R signals are inactive.

30.2.5 Auto clear register. The 8-bit auto clear register contains one bit for each IREQ input and specifies the operating mode for each of the ISR bits. When an auto clear bit is off, the corresponding ISR bit is set when that interrupt is acknowledged and is cleared by software command. When an auto clear bit is on, the corresponding ISR bit is cleared by the hardware at the end of the acknowlege sequence. A reset function clears all auto clear bits.

MODULE CONTROL INPUT							DATA BUS OPERATION
-DS-W	A2	-CS	A1	READ	-DS-R	-IACK	
1	0/1	0	0	1	0	1	Transfer contents of AM9519A unit 1/2 preselected data register data bus
0	0/1	0	0	0	X	1	Transfer contents of data bus to AM9519A unit 1/2 preselected data register
1	0/1	0	1	1	0	1	Transfer contents of status register to data bus
0	0/1	0	1	0	X	1	Transfer contents of AM9519A unit 1/2 data bus to command data bus to command register
X	0/1	1	X	X	0	0	Transfer contents of AM9519A unit 1/2 selected response memory location to data bus
X	0/1	1	X	X	X	1	No information transferred

FIGURE 3. Summary of data bus transfers.

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30.2.6 Status register. The 8-bit status register contains information concerning the internal state of the chip. It is especially useful when operating in the polled mode in order to identify interrupting devices. Figure 4 shows the status register bit assignments. Bits S0-S2 are set asynchronously to a status register read operation. It is recommended to read the register twice and to compare binary vectors for equality prior to proceeding with the device service in polled mode. The polarity of the GINT bit 7 is not affected by GINT polarity control (mode bit 3). The status register is read by executing a read operation (-CS=0, A2=0/1, READ=1, -DS-R=0, -DS-W=1) with the control location selected (A1=1).

30.2.7 Mode register. The 8-bit mode register controls the operating options of the AM9519A. Figure 5 shows the bit assignments for the mode register. The five lowest order mode bits (0 through 4) are loaded in parallel by command. Bits 5, 6 and 7 are controlled by separate commands. (See figure 2). The mode register cannot be read directly to the data bus, but mode bits 0, 2 and 7 are available as part of the status register.

30.2.7.1 Mode bit 0. Mode bit 0 specifies the rotating/fixed priority mode. In the fixed mode, priority is assigned to the request inputs based upon their physical location at the chip interface, with IREQ0 the highest and IREQ7 the lowest for each AM9519A. In the rotating mode, relative priority is the same and the most recently serviced request is assigned the lowest priority.

30.2.7.2 Mode bit 1. Mode bit 1 selects the individual/common vector option. Individual vectoring provides a unique location in the response memory for each interrupt request. The common vector option always supplies the response associated with IREQ0 no matter which request is being acknowledged.

30.2.7.3 Mode bit 2. Mode bit 2 specifies interrupt or polled operation. In the polled mode the group interrupt output is disabled. The microprocessor may read the status register to determine if a request is pending. Since the -IACK pulses are not normally supplied in polled mode, the IRR bit is not automatically cleared, but may be cleared by command. With no -IACK input, the ISR and response memory are not used.

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30.2.7.4 Mode bit 3. Mode bit 3 specifies the sense of the GINT output. When active high polarity is selected, the output is a two-state configuration. For active low polarity, the output is open drain and requires a pull-up resistor to provide the high logic level. The active low polarity must be chosen on the SQY because the individual GINT outputs are wire-ORed on the module.

30.2.7.5 Mode bit 4. Mode bit 4 specifies the sense of the IREQ inputs. When active low polarity is selected, the IRR responds to falling edges on the request inputs. When active high is selected, the IRR responds to rising edges.

30.2.7.6 Mode bits 5 and 6. Mode bits 5 and 6 specify the register that will be read on subsequent data read operations. This preselection remains valid until changed by a reset or a command.

30.2.7.7 Mode bit 7. Mode bit 7 is the master mask bit that disables all request inputs. It is used to disable all interrupts without modifying the IMR so that previous IMR contents are valid when interrupts are re-enabled.

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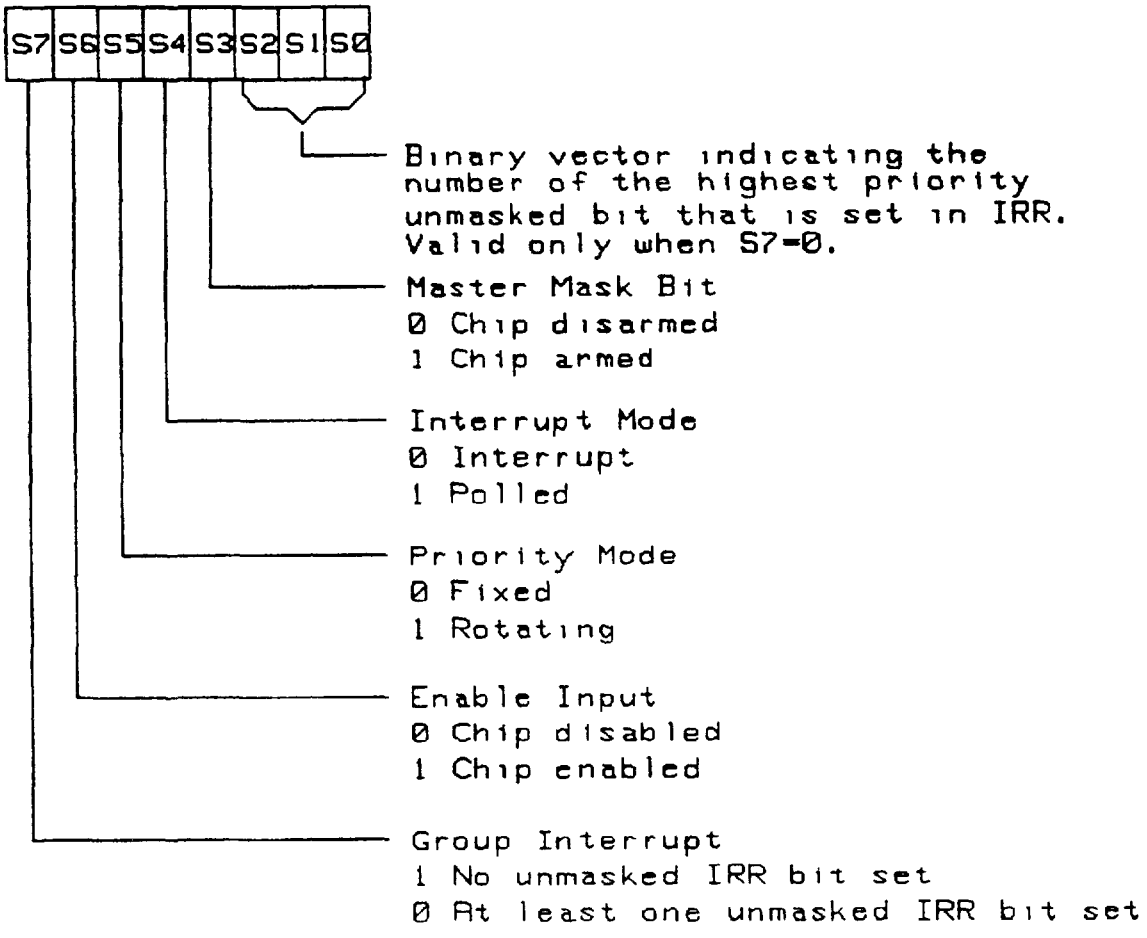


FIGURE 4. Status register bit assignments.

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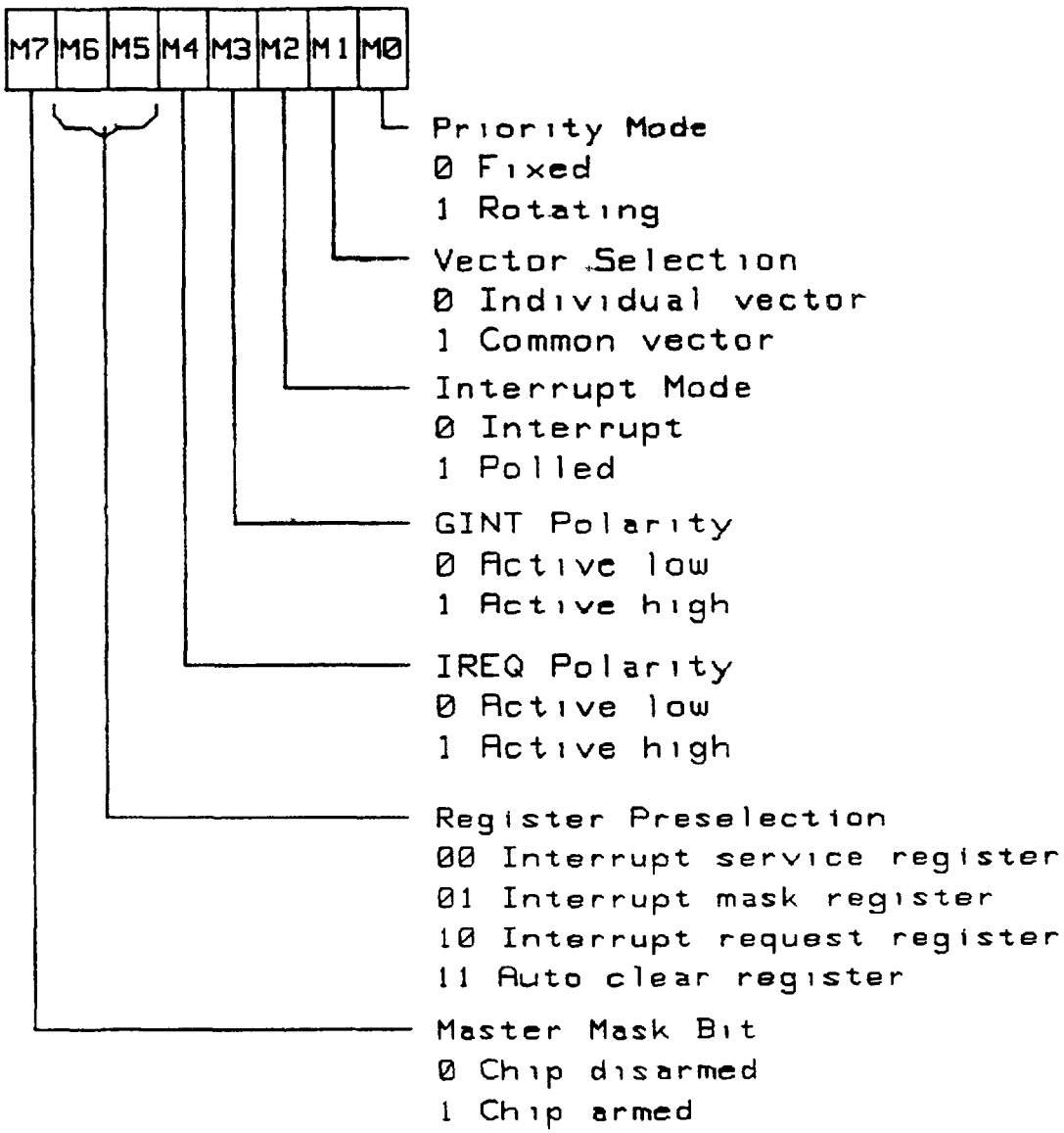


FIGURE 5. Mode register bit assignment.

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30.2.8 Command register: The 8-bit command register stores the last command entered. Depending upon the command opcode, it may initiate internal actions or precondition the part for subsequent data bus transfers. The command register is loaded by executing a write operation (-DS-W=0, READ=0, -CS=0, A2=0/1) with the control location selected (A1=1) as shown in figure 2.

30.2.9 Byte count register: The length in bytes of the response associated with each interrupt is independently programmed so that different interrupts may have different length responses. The byte count for each response is stored in eight 2-bit byte count registers. For a given interrupt, the AM9519A will expect to receive a number of -IACK pulses that equals the corresponding byte count, and will hold -RIP low until the count is satisfied. Figure 6 shows the coding assignments for the byte count register.

30.3 Initialization. After reset, the AM9519A must be initialized by the microprocessor in order to perform useful work. At a minimum, the master mask bit and at least one of the IMR bits should be enabled. If vectoring is to be used, the response memory must be loaded; if not, the mode must be changed to a non-vectored configuration. Normally, the first step will be to modify the mode register and the auto clear register in order to establish the configuration desired for the application. Then the response memory and byte count will be loaded for those request levels that will be in use. Finally, the master mask bit and at least portions of the IMR will be enabled to allow interrupt processing to proceed.

BY1	BY0	COUNT
0	0	1
0	1	2
1	0	3
1	1	4

FIGURE 6. Byte count coding.

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3 DOCUMENT TITLE
Modules, Digital, Standard Electronic Interrupt Control Unit Module, Key Code SQY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

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